Model Question Paper Reg.No :……

Name:……

MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS)
EXAMINATION October 2026
V SEMESTER
Major- MG5DSEPHY304- PHYSICS OF ADVANCED MATERIALS
(2024 ADMISSION ONWARDS)

Duration: 2 hours Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions
Answer any 10 questions
Each question carries 2 marks

1. What are glass-ceramics? [U/CO1]
2. Give two examples of elastomers and mention one important application of each. [K/CO1]
3. Differentiate between Single-walled CNT and multi-walled CNT [K/CO1]
4. Name the three typical types of stress–strain behaviour exhibited by polymeric materials. [K/CO1]
5. What is viscoelastic creep? [U/CO1]
6. What is matrix phase in a composite? [U/CO1]
7. Write the relation between drift velocity and mobility. [K/CO2]
8. Define Hall voltage. [K/CO2]
9. What are conducting polymers? Explain the role of doping in enhancing their electrical conductivity, with suitable examples. [K/CO2]
10. Explain why repeatedly dropping a permanent magnet on the floor causes it to become demagnetized. [U/CO3]
11. Write the mathematical expression for steady-state heat conduction [K/CO4]
12. Why gold appear yellow in color?. [U/CO5]

(10x2=20)